

Answer all questions.

1. (a) Salts can be produced by reacting acids with alkalis.

(i) Complete the following equation for the reaction between an acid and alkali. [1]



(ii) **Circle** the word which best describes the reaction between an acid and an alkali. [1]

displacement

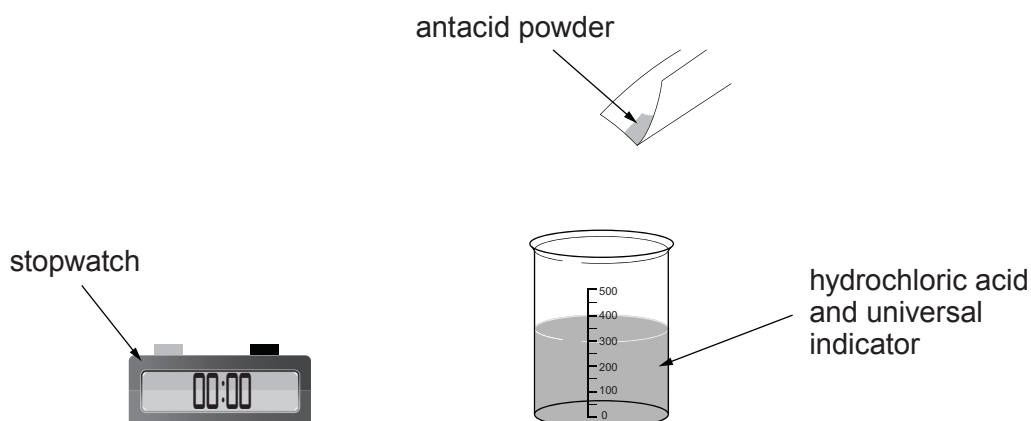
neutralisation

oxidation

reduction

(b) Indigestion can be caused by excess hydrochloric acid in the stomach. To treat indigestion, antacid powders are commonly used.

A group of pupils used the following apparatus to compare three brands of antacid powder, to see which was the most effective at treating acid indigestion.



They added an equal mass of each of the antacid powders to separate beakers, containing equal amounts of hydrochloric acid and universal indicator.

They stirred the mixture and recorded the time taken for the universal indicator to turn green in each beaker. They carried out the test three times for each antacid powder. Their results are shown in the table.



Examiner
only

Antacid powder	Time taken for the universal indicator to turn green (min : s)			
	Result 1	Result 2	Result 3	Mean
Brand 1	5 : 25	5 : 36	5 : 14	5 : 25
Brand 2	4 : 28	3 : 20	4 : 32	4 : 30
Brand 3	2 : 28	2 : 30	2 : 44	2 : 34

(i) State which **two** results were used to calculate the mean value for brand 2. [1]
..... and

(ii) Convert the mean time for brand 2 into **seconds**. [1]

Mean time = s

(iii) Give the reason why the results suggest that brand 3 is the best powder for treating acid indigestion. [1]

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